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14 June 1985
SUPERSEDING
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MILITARY SPECIFICATION

INSULATING AND JACKETING COMPOUNDS, ELECTRICAL (FOR CABLES, CORDS, AND WIRES), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general composition and the physical, chemical, and low temperature properties of certain polymeric compounds used in the insulating and jacketing of electrical cables, cords, and wires, as determined on specimens removed therefrom, but does not cover compounds for hookup wires nor items where the wall thickness of the insulation or jacket is less than .006 inch. It does not cover characteristics associated with the finished product, except that test specimens are taken from the finished product. This specification is intended to serve as a auxiliary to detailed specifications for finished cables, cords, or wires (see 6.2 and 6.3).

1.2 Classification. Insulating and Jacket Compounds shall be of the following types as specified on the applicable specification sheets (see 6.3).

INSULATING COMPOUNDS

Type IP - Medium-low temperature polyvinylchloride (PVC) and polyvinylchloride acetate (PVCA).
Type IS - Medium-low temperature, styrene-butadiene rubber (SBR).
Type IS-L - Medium-Low temperature, heat-resistant styrene-butadiene rubber (SBR).
Type IR - Low temperature natural rubber.1/

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type IL - Low temperature natural rubber deposited from a rubber-latex compound. 1/, 2/

Type IJ-S - Medium-low temperature insulating and jacketing styrene-butadiene rubber (SBR).

Type IL-H - Low temperature heat-resistant natural rubber deposited from a rubber-latex compound. 1/, 2/

Type IL-RN - Low temperature layer of natural rubber latex and a color-coded outer layer of chloroprene rubber (CR) latex bonded together.

Type IH - Low temperature heat and ozone resistant ethylene-propylene rubber (EPM) or ethylene-propylene-diene rubber (EPDM). 3/

Type IJ-H - Low temperature, heat and weather resistant insulating and jacketing ethylene-propylene rubber (EPM) or ethylene-propylene-diene rubber (EPDM). 3/

Type IE - Polyethylene in accordance with L-P-390, Type II, Class L, Grade 3 or 4, Category 5.

Type TPE-S - Low temperature, heat and ozone resistant, styrenic thermoplastic elastomer.

Type TPE-O - Low temperature, weather resistant, olefinic thermaplastic elastomer.

JACKETING COMPOUNDS

Type JP - Medium-low temperature polyvinylchloride (PVC) and polyvinylchloride-acetate (PVCA).

Type JS - Medium-low temperature syrene-butadiene rubber (SBR).

Type JS-L - Low temperature heat-resistant styrene-butadiene rubber.

Type JR - Low temperature natural rubber. 1/

Type JN - Moderate-low temperature chloroprene rubber (CR).

Type JN-L - Low temperature chloroprene rubber (CR).

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Type JH - Low temperature, heat and weather resistant ethylene-propylene (EPM) or ethylene-propylene-diene rubber (EPDM). 3/

Type JU - Low temperature, heat and weather resistant polyurethane thermoplastic elastomer.

1/ Synthetic polyisoprene may be blended with natural rubber.

2/ Hereinafter referred to as rubber latex.

3/ EPM or EPDM rubber may be blended with any of the other elastomers except for Types IP, IE, JP and JU. Compounds of resulting blends, with a maximum of 75 percent by weight, shall meet the requirements of the Type specified, except compounds containing over 75 percent by weight shall be identified as EPM or EPDM. When EPM or EPDM is at least 35% by weight of total polymer blend, the compound need not meet the requirement for addition of ozone protective waxes of 3.3.1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390 - Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density).

STANDARDS

FEDERAL

Test Method Standard No. 228 - Cable and Wire, Insulated; Methods of Testing

Test Method Standard No. 601 - Rubber; Sampling and Testing

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-45662 - Calibration Systems Requirements
(See Supplement 1 for list of associated specification sheets)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activitiy or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D518 - Rubber Deterioration - Surface Cracking, Standard Test Method for

ASTM D746 - Brittleness Temperature of Plastics and Elastomers by Impact, Standard Test Method for

ASTM D1053 - Measuring Rubber Property - Stiffening at low Temperature using a Torsional Wire Apparatus, Standard Method for

ASTM D1149 - Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen), Standard Test Method for

ASTM D1765 - Carbon Blacks used in Rubber Products, Standard Classification System for

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 First article. Insulating and jacketing materials to be furnished under this specification shall be a product which has passed the first article inspection specified in 4.4.

3.3 Materials. Unless otherwise specified (see 3.1) the materials listed shall be as specified herein. Prior approval to use substitute material must be obtained from the qualifying activity. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Ozone-protective wax. Natural rubber, styrene-butadiene rubber (SBR), and chloroprene rubber (CR) compounds (except latices of these materials) shall incorporate an "ozone protective" wax to protect the insulation and jacket against cracking due to exposure.

3.3.2 Insulating compounds. Natural rubber (except latices of this material), and styrene-butadiene rubber (SBR) insulating compounds shall contain not less than 2.5 parts by weight of ozone protective wax per 100 parts by weight of base rubber polymer.

3.3.3 Jacketing compounds.

3.3.3.1 Natural rubber and styrene-butadien rubber (SBR). Natural rubber and styrene-butadiene rubber (SBR) jacketing compounds shall contain not less than 4 parts by weight of ozone protective wax per 100 parts by weight of base rubber-polymer.

3.3.3.2 Chloroprene rubber (CR). Chloroprene rubber (CR) jacketing compounds shall contain not less than 3 parts by weight of ozone protective wax per 100 parts by weight of base rubber-polymer.

3.3.3.3 Polyurethane elastomer. Polyurethane elastomer shall contain an ultraviolet inhibiting pigment of furnace or channel black of particle size averaging 20 nm or less in diameter, in the amount of .5 to .7 parts by hundred (by weight) of elastomer suitable furnace blacks are identified by the designation N 110 in accordance with ASTM D1765.

3.3.4 Free elemental sulfur. Rubber latex and rubber latex-chloroprene rubber latex compounds shall not contain more than .5 percent free elemental sulfur, natural rubber compounds not move than .1 percent free elemental sulfur,and styrene-butadiene rubber compounds and ethylene-propylene-diene rubber compounds not more than .25 percent free elemental sulfur. (see 4.6.1 and 6.5).

3.4. Ozone resistance. After testing in accordance with 4.13, the rubber test specimem shall exhibit no visible cracks when examined under a seven power optical magnifier.

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3.5 Properties of insulating and jacketing. The insulating and jacketing materials furnished under this specification shall conform to the requirements in the individual specification sheet. Conformance shall be determined by subjecting specimens of the insulating and jacketing material (removed from cables and cords) to the physical property requirements in the applicable specification sheet.

3.6 Workmanship. Insulating and jacketing material shall be uniform in quality and shall be free from any burrs, die marks, chatter marks, foreign material and other defects that will affect life, serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be in accordance with standards established by the contractor for calibrating the measuring and test equipment used in controlling product quality. Calibration of the standards which control the accuracy of inspection equipment shall comply with the requirements of MIL-STD-45662.

4.2 Classification of inspections. The examinations and tests of insulating and jacketing materials are classified as follows:

a. Materials inspection. (see 4.2.1).
b. First article inspection (see 4.4).
c. Quality conformance inspection. (see 4.5).

(1) Inspection of product for delivery. (see 4.5.1).
(2) Inspection of preparation for delivery. (see 5).

4.2.1 Materials inspection. For all cables, cords, or wires submitted to the Government for acceptance, the contractor shall furnish certification of compliance from the source of manufacture for each batch of compound used in the composition of the insulation and jacket. This certification shall indicate compliance with 3.3 to 3.3.3.3 inclusive (see 6.3).

4.3 Inspection conditions. Unless otherwise specified in the specific cable, cord or wire specification referencing this specification, all tests shall be conducted in a standard atmosphere as specified in section 6 of Federal Test Method Standard No. 228.

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4.4 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article inspection is valid only on the contract under which it is granted, unless extended by the Government to another contract (see 3.1 and 6.7).

4.4.1 Sample size. Provide six inch lengths of Cable, Cord or Wire sample units required to perform the tests (see 4.4.2) for first article inspection which shall be submitted in accordance with MIL-STD-105, unless otherwise specified (see 6.3).

4.4.2 Test routine. Sample units shall be subjected to the inspections specified in 3.3.4, 3.4 and 3.5.

4.4.3 Submission of data for approval. Test results shall be forwarded to the contracting officer for approval (see 6.3).

4.4.4 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection. Quality conformance inspection shall be in accordance with Table I.

TABLE I. QUALITY CONFORMANCE INSPECTION.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
Group A		
Free elemental sulfur	3.3.4	4.6.1
Tensile strength	3.5	4.6.2.2
Elongation	3.5	4.6.2.3
Set	3.5	4.6.2.4
Tensile stress	3.5	4.6.2.5
Tear resistance	3.5	4.6.2.6
Oxygen-bomb aging	3.5	4.6.2.7
Air-pressure aging	3.5	4.6.2.8
Air-oven aging	3.5	4.6.2.9
Oil immersion	3.5	4.6.2.10
Durometer hardness	3.5	4.6.2.11
Cold-tension recovery	3.5	4.10
Shrink-back	3.5	4.14
Group B		
Brittleness	3.5	4.7
Torsional-stiffness ratio and torsional modulus	3.5	4.9
Group C		
Ozone reistance	3.4	4.13
Hydrolytic stability	3.5	4.15

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4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Groups A, B, and C inspection. Delivery of products which have passed the groups A and B inspection shall not be delayed pending the results of the Group C inspection.

4.5.1.1 Inspection lot. An inspection lot shall be as defined in the end item specification. When not defined therein, it shall consist of all the sample units (lengths, reels, spools, coils) of cables, cords, or wires of a specific type and AWG size, the compounds for which were made under substantially the same conditions, from the same batch, and offered for inspection at one time.

4.5.1.2 Unit of product. When not defined in the end item specification, the unit of product shall be taken as 1,000 feet of cables, cords or wires, whether that length be part of the spool, reel, or coil or consist of one or more spools, reels, or coils.

4.5.1.3 Sample. The sample shall consist of the number of units of product required by the sampling plan for the size of the lot submitted.

4.5.1.4 Selection of sample units. The number of sample units required for inspection shall be chosen by selecting one spool, reel, or coil for each sample unit. Samples shall be selected at random intervals throughout the lot. After which the selected spool, reel, or coil shall be treated as the sample unit for purposes of inspection.

4.5.1.4.1 Molded sheets. Molded sheets of .075 to .085 inch thick may be furnished for certain tests of small size cables when authorized in 4.6.2.6., The stock of these sheets shall be taken from the same batch as the insulation or jacket that they represent (preferably uncured rubber for these sheets will be taken from the feed to the extruder and record kept of the cable produced from that run). The sheets shall be at the same state of cure as the insulation or jacket that they represent. The state of cure shall be considered to be the same if the tensile stress of the sheet at 200 percent elongation is within 20 percent of that of the insulation or jacket it represents. The sheets shall be marked to show the direction of calendar or mill grain.

4.5.2 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. From each sample unit (each length, reel, spool, or coil), a sample length of cable, cord, or wire shall be cut. Specimens as specified in 4.6.2.1, 4.6.2.5 and 4.6.2.6 and 4.6.3.3 shall be cut from each sample length. The reduced inspection procedure shall be R-1. The acceptable quality levels (AQL) shall be 6.5 (percent defective) and the inspection level shall be S-4 for normal inspection and S-3 for reduced inspection. Determination of whether or not the individual sample unit is defective in a particular test shall be made as specified in the applicable method of test. (See 4.6.2) The specimens for each color of insulation shall constitute a separate determination, and the comparison of aged and unaged values shall be made color for color. If a sample unit is found defective in more than one test, it shall be considered as one defective unit. The total number of defective units shall be compared with the acceptance number for that sample size. If the acceptance number is exceeded, the lot shall be rejected.

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4.5.2.2 Sampling plan for group C. Sufficient length of cable, cord or wire required to perform tests in Table I shall be selected from a 5000 feet length of cable, cord or wire, unless length other than 5000 feet is specified in the end-item specification or the amount produced in six months, whichever is less frequent.

4.5.2.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken Group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful.In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6 Methods of test.

4.6.1 Free elemental sulfur. Free elemental sulfur content shall be determined in accordance with method 16211 of Federal Test Method Standard No. 601, except that where greater sensitivity is required, a .01 iodine solution shall be used and the blue end point shall last 3 to 4 seconds.

4.6.2 Physical properties.

4.6.2.1 Specimens. Specimens shall be taken from each sample length (see 4.5.1.4). The number of specimens specified in the applicable test method shall be prepared as described therein from the jacket and from the insulation of each conductor. For small runs of cable or cord where only one basic solid color insulation is used and coding is accomplished by striping, lettering, numbering, or a non-adhering tape or braid, each conductor need not be tested, but at least one shielded conductor and one unshielded conductor (as applicable) from each layer of the cable core shall be tested; the applicability of this provision for reduced number of conductors to be tested shall be as determined by the contracting officer. The shield shall be removed from the shielded conductor prior to subjection to test conditions and tests.

4.6.2.1.1 Size of specimen for test of tensile properties. In the case of conductor smaller than AWG size 6, the insulation test specimen may be an entire section of the insulation, 6.000 inch long. When the full cross section is used, the specimen shall not be cut longitudinally unless necessary to remove the conductor. In the case of AWG size 6 or larger, 6.000 inch-long specimens, approximately square in section with cross section not greater than .025 square inch, shall be cut from the insulation. In extreme cases it may be necessary to use segmented or sector-shaped specimens. The preferred specimen of the jacket is a 6.000 inch length with a die-cut reduced section 2.313 inches long by .250 inches wide. Gage marks shall be applied before conditioning.

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4.6.2.1.2 Condition and age. Specimens shall not be heated, immersed in water, or subjected to any mechanical or chemical treatment not specified herein. No tests shall be made within 24 hours after manufacture unless agreed to by the manufacturer.

4.6.2.1.3 Calculation of percent change in values. The calculation of percent change in tensile strength, elongation, etc., after specified conditioning or aging period shall be made for each individual specimen. When unaged specimens are destroyed during testing, the basis for percent change shall be the arithmetic mean (average) value of the unaged specimens which meet the qualifying value.

4.6.2.2 Tensile strength. Tensile strength shall be determined in accordance with method 3021 of Federal Test Method Standard No. 228, except that flat, smooth, steel grips of the self-aligning type may be used.

4.6.2.3 Elongation. Elongation shall be determined in accordance with method 3031 of Federal Test Method Standard No. 228, except that bench marks shall be 2 inches apart, and flat, smooth, steel grips of the self-aligning type may be used. The elongation after aging shall be based on the bench marks applied before aging.

4.6.2.4 Set. Set shall be determined in accordance with method 3161 of Federal Test Method Standard No. 228, except that:

a. Bench marks shall be 2 inches apart.
b. The test specimens shall be stretched so that the distance between bench marks is 6 inches; they shall then be released within 5 seconds.
c. Set shall be determined 1 minute after the beginning of release.
d. Flat, smooth, steel grips of the self-aligning type may be used.

4.6.2.5 Tensile stress. Apparatus and procedure shall be as specified in method 3041 of Federal Test Method Standard No. 228.

4.6.2.6 Tear resistance. Apparatus and procedure shall be as specified in method 3111 of Federal Test Method Standard No. 228. The specimens shall be cut and buffed from a section of the jacket under test except as indicated in 4.6.2.6.1.

4.6.2.6.1 Thin-walled jackets. For tear resistance tests of jackets having an allowed minimum thickness of less than .040 inch, specimens may be cut from sheets specified in 4.5.1.4.1. Specimens shall be cut so that tear will occur in the direction of the calender or mill grain.

4.6.2.7 Oxygen-bomb aging. Specimens, after being subjected to an oxygen-bomb aging specified in method 4011 of Federal Test Method Standard No. 228, under the required conditions of time and temperature specified in the applicable specification sheet shall be subjected to the tests specified in 4.6.2.2 and 4.6.2.3. Specimens of insulation shall remain on the wire during aging.

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4.6.2.8 Air pressure aging. Specimens, after being subjected to the air-pressure aging specified in method 4021 of Federal Test Method Standard No. 228; under the required conditions of time and temperature specified in the applicable specification sheet shall be subjected to the tests specified in 4.6.2.2 and 4.6.2.3. Specimens of insulation shall remain on the wire during aging.

4.6.2.9 Air-oven aging. Specimens, after being subjected to the air-oven aging specified in method 4031 of Federal Test Method Standard No. 228 under the required conditions of time and temperature specified in the applicable specification sheet, shall be subjected to the tests specified in 4.6.2.2 and 4.6.2.3. Specimens of insulation shall remain on the wire during aging.

4.6.2.10 Oil immersion. Oil immersion shall be performed in accordance with method 4223 of Federal Test Method Standard No. 228, except that for type JP (polyvinylchloride and polyvinylchloride-acetate (PVC), the temperature shall be $70^{\circ} \pm 1^{\circ}C$, the immersion period 18 hours, and the recovery time 16 to 48 hours.

4.6.2.11 Durometer hardness. Durometer hardness shall be determined in accordance with method 3021 of Federal Test Method Standard No 601.

4.6.3 Low temperature tests.

4.6.3.1 General procedure. Subsequent to insertion in the low-temperature cabinet, test specimens shall be handled only with tongs which have been conditioned at the test temperature for at least 2 hours. Test tongs shall be handled with gloved hands after gloves have been conditioned for at least 5 minutes. Subsequent to any handling of a test specimen, at least 30 seconds shall elapse prior to testing. Care shall be taken to maintain the specified temperature (see 3.1) during the conditioning and test periods while loading with dry ice, opening the handholes, etc.

4.6.3.2 Low-temperature cabinet. The low-temperature cabinet to be used for conditioning specimens in determining low-temperature properties shall be of the mechanically refrigerated or dry-ice type, and shall conform to the following requirements:

a. The heat-transfer medium in the test chamber shall be air or a mixture of air and carbon dioxide.

b. Temperatures shall be maintained at $-55 \pm 1^{\circ}C$ ($-67 \pm 2^{\circ}F$) or -40 ($\pm 1^{\circ}C$ or $\pm 2^{\circ}F$) as specified on the applicable specification sheet.

c. The heat-transfer medium shall be thoroughly circulated in the test chamber by means of mechanical agitation, which shall function at all times during both the conditioning and testing periods.

d. Automatic temperature control shall be used.

e. Each of the dimensions of the test chamber interior shall be at least 24 inches.

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f. If incandescent lamps are provided for illumination, they shall be located at least 16 inches from the test specimens.

g. For cabinets other than those of the walk-in type, the following shall apply:

(1) At least one observation window shall be provided in one side of the cabinet. Each window shall be either 3 or 5-ply glass construction. Each window shall be at least 7.000 inch square.

(2) For operation of the test equipment, handholes equipped with insulated sleeves shall be installed in the wall of the cabinet. Handholes shall be at least 5.000 inch in diameter.

(3) In the dry-ice type of cabinet, a fan or blower in the dry-ice compartment shall circulate the heat-transfer medium from the dry-ice compartment into the test chamber, and back. The amount and movement of the heat-transfer medium shall be adjusted to minimize the fluctuations of the heat in the test chamber.

4.6.3.3 Test specimens. Unless otherwise specifically authorized, test specimens for brittleness, torsional-stiffness ratio, and torsional-modulus and cold tension-recovery tests, shall be cut from samples of cord or cable to obtain specimens .080 inch wide, 1.500 inches long with .250 inch square stubs at each end (T-50 specimens). The thickness of specimens shall be .075 inch (+.010 or -.035 inch). The specimens shall be cut with a sharp die.

4.6.3.3.1 Torsional-stiffness ratio. If the angle of twist of the specimen described in 4.6.3.3 is 170° or more at normal ambient temperature, one of the following specimens shall be used in lieu of the regular T-50 specimen:

a. A tube specimen as specified in method 3021 of Federal Test Method 228 for insulation on conductors AWG sizes 16 and 18.

b. A modified T-50 specimen .125 inch wide by 1.500 inch long with .250 inch-square stub for insulation on all other conductor sizes, and for all jacket stocks.

c. Specimens prescribed in 4.6.3.3 cut from sheets specified in 4.5.1.4.1 may be used in lieu of specimens cut from insulation or jackets if the minimum allowed thickness of the insulation of jacket is less than .040 inch.

4.7 Brittleness.

4.7.1 Apparatus. The test apparatus shall conform to Method 5311 or Method 5321 of Federal Test Method Standard No. 601, with air as the heat transfer medium. If the instrument specified in Method 5321 is used, with air as the heat transfer medium. If the instrument specified in Method 5321 is used, no more than one specimen shall be tested at one time.

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4.8 Specimens.

4.8.1 Description. Specimens shall be as prescribed by 4.6.3.3 with exceptions allowed in 4.8.2.

4.8.2 Specimens allowed by 4.6.3.3.1 may be used under the conditions imposed by 4.6.9.2a, b, and c.

4.8.3 Specimens conforming to ASTM D746 may be used, if the supplier desires.

4.8.4 Number of specimens. Four jacket specimens and four specimens of each color of insulation shall be cut from the sample unit.

4.8.5 Procedure. Two specimens shall be air-oven aged for a minimum of 48 hours at 160 ± 2°F and allowed to cool at room temperature prior to conditioning at sub-zero temperatures. The aged and unaged specimens shall be conditioned in the low temperature cabinet for a period of 7 days at a temperature of -55 ± 1°C (-67 ± 2°F). The test apparatus shall be conditioned for not less than 2 hours at the applicable temperature and the test shall be conducted while specimens and tester are still at the same temperature. After the impact blow is delivered, the specimens shall be examined. Failure shall be considered to have occurred if fractures or cracks result.

4.9. Torsional-stiffness ratio and torsional modulus.

4.9.1 Apparatus. The test instrument shall be a Gehman torsional tester as described in Method 5612 of Federal Test Method Standard No. 601, except that only the torsion apparatus, the stand, and a suitable bracket to hold the lower clamp are required. A beryllium-copper-type torsion wire of appropriate torsion constant shall be used. In cases where two or more Gehman torsional testers are used (one at normal temperature and one at low temperature), the torsion wires shall be calibrated as described in Method 5612 of Federal Test Method Standard No. 601. The wires shall be considered suitable only if their torsional constants are within ± 3 percent of their specified values.

4.9.2 Torsional-stiffness ratio. If the angle of twist of the specimen described in 4.6.3.3 is 170° or more at normal ambient temperature, one of the following specimens shall be used in lieu of the regular T-50 specimen:

a. A tube specimen as specified in Method 3021 of Federal Test Method Standard No. 228 for insulation on conductors AWG sizes 16 and 18.

b. A modified T-50 specimen .125 inch wide by 1.500 inch long with .250 inch-square stub for insulation on all other conductor sizes, and for all jacket stocks.

c. Specimens prescribed in 4.6.3.3 cut from sheets specified in 4.5.1.4.1 may be used in lieu of specimens cut from insulation or jackets if the minimum allowed thickness of the insulation or jacket is less than .040 inch.

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4.9.3 Number of specimens. Three jacket specimens and three specimens of each type of insulation shall be cut from the sample unit (see 4.6.2.1 and 4.9.2 (c))

4.9.4 Procedure. The test procedure shall be as follows:

a. With the specimens and the test instrument at a temperature ranging from 20°C (68°F) to 26°C (78°F), the lower end of a specimen shall be secured in the rigidly mounted clamp of the test instrument. The upper end is secured in the upper pinch clamp attached to the lower end of the torsional wire.

b. The pointer shall be adjusted to the 0 position by rotating the protractor scale. The torsional head shall be turned 180° in 1 second or less, and the pointer reading shall be recorded after a period of 10 seconds.

c. After conducting the above tests, the specimens shall be conditioned in the low-temperature chamber for a period of 7 days at a temperature of -55°C ±1°C. The test apparatus shall be conditioned for not less than 2 hours at the same temperature.

d. The test procedure in (a) and (b) above shall be repeated on the specimens at the end of the 7-day conditioning period. The torsional-stiffness ratio for each specimen shall be calculated.

e. When applicable, measurements shall be made as required to calculate the torsional modulus in pounds per square inch.

4.9.5 Calculations. The torsional-stiffness ratio shall be calculated as follows:

$$TSR\ -55 = TSF\ -55/TSF\ 23^{\circ}C$$

Where TSR -55 = torsional-stiffness ratio at -55°C

TSF -55 = torsional-stiffness factor at -55°C

TSF 23 = torsional-stiffness factor at 23°C

$$\text{The torsional-stiffness factor} = \frac{180^{\circ} - (\text{ANGLE OF TWIST OF SPECIMEN})}{\text{ANGLE OF TWIST OF SPECIMEN}}$$

4.9.6 Torsional modulus. The torsional modulus at -55°C (-67°F) shall be calculated in accordance with paragraph 5 of Method 5612 of Federal Test Method Standard No. 601. Torsional stiffness factors may be referenced in accordance with Table I of ANSI/ASTM D1053.

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4.10. Cold-tension recovery.

4.10.1 Apparatus. The test instrument shall consist of a measuring board on which several stretching devices can be mounted. Each stretching device shall have one movable clamp and one fixed clamp. The board shall have lines engraved at intervals corresponding to each 10-percent stretch, based on the length of the specimen between the .250 inch-square stubs.

4.11. Specimens.

4.11.1 Description. Specimens shall be as described in 4.6.3.3 with exceptions allowed by 4.6.3.3.1 under the conditions imposed by 4.6.3.3.1a, b, and c. If the specimens specified in 4.6.3(a) are used, bench marks shall be placed on each specimen, 1.500 inches apart, to be used as reference for extension and recovery measurements.

4.11.2 Number of specimens. Three jacket specimens and three specimens of each type of insulation shall be cut from the sample unit (see 4.6.2.1).

4.12. Procedure.

a. With the specimens at temperatures specified in 4.12(b), they shall be clamped in the stretching devices along the edges of the studs adjacent to the linear part of the specimen and laid on the board. The movable clamp shall be pulled back until the 1.500 inch-long portion of the specimen has been stretched to 100-percent elongation, and fixed in that position.

b. The stretching devices and the specimens shall be conditioned in the low-temperature chamber for 7 days at the temperature specified in the applicable specification sheet. The measuring board shall be conditioned for not less than 2 hours at the same temperature.

c. With the test instrument and specimens still in the low-temperature chamber, the movable clamp shall be released from its fixed position, and the assembly shall be conditioned for an additional period of 30 minutes at the low temperature specified in 4.12(b).

d. The final length of the specimen shall be determined with the board stretching devices, and specimens held at an angle of 15° from the vertical.

e. The cold-tension-recovery percentage for each specimen shall be calculated.

4.12.1 Calculations. The percentage of cold-tension-recovery shall be computed from the formula:

$$\text{Percentage cold-tension-recovery} = \frac{(L_S - L_F)}{(L_S - L_O)} \times 100$$

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Where:

L_C = stretched length of specimen

L_F = final length of specimen

L_O = initial length of specimen

4.13 Ozone resistance test.

4.13.1 General procedure. The tests for ozone resistance shall be performed in accordance with the procedures prescribed in ANSI/ASTM D1149 except as indicated below for ozone concentration and test temperature.

4.13.2 Test specimens. The test specimen shall be a length of completed cable for jacket evaluation or a length of insulated wire taken from a completed cable. Not less than six specimens each of jacketed cable and six specimens each of all color insulation shall be tested (see 4.6.2.1).

4.13.3 Preparation. The length of the specimen shall be sufficient for at least five turns on the appropriate mandrel as listed below plus fastening to the mandrel; three turns may be used for cables greater than one inch in diameter. Shielding, separators, and braid shall be removed from the insulated wire specimens.

4.13.4 Conditioning of specimens. Samples of the cable from which specimens are to be taken shall be maintained at room temperature for at least one week prior to preparation of specimens. One set of specimens (minimum of three specimens of jacket and three specimens of each color insulation) shall then be air-oven aged in an unstressed condition for 70 ±1/2 hours at 158°F (70°C) and then kept at room temperature at least four days prior to ozone exposure. The second set (consisting of a minimum of three each jacket and three each insulation of each color) shall be exposed to ozone as required without oven conditioning. Each specimen shall be wrapped around a mandrel of diameter indicated below and exposed thus in the ozone chamber.

SPECIMEN DIAMETER (INCH)	MANDREL DIAMETER
Up to .500	4 X cable OD
.501 to .750	5 X cable OD
.751 to 1.250	6 X cable OD
1.251 to 1.750	8 X cable OD
1.751 and over	10 X cable OD

4.13.5 Chamber environment. The ozone concentration used in this test shall be 50 ±3 parts of ozone per 100,000,000 parts of air. The temperature shall be as specified in the applicable specification sheet.

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4.13.6 Procedures. Specimens shall be exposed for a period of 168 ±1 hours. At the end of the exposure time, the mounted specimens shall be removed from the chamber and allowed to cool at room temperature. While on the mandrel, the specimens shall be examined for compliance with 3.3.

4.14 Shrink-back. A 7.000 inch long specimen shall be cut from the insulated conductor. Shield and/or color coding braid, tape, or jacket shall be removed. The specimen shall then be cut to 6.000 inch with conductor flush with the insulation. The specimen shall then be air-oven aged at 99±1°C for 24 ±1/2 hours. The specimen shall then be removed and allowed to cool to room temperature. The amount the conductor extends beyond the insulation shall be measured at each end, and the insulation shall be examined for splitting or cracking. The amount of insulation shrinkback at either end shall be no greater than specified in the applicable specification sheet and there shall be no evidence of splitting or cracking.

4.15 Hydrolytic stability for Type JU, Polyurethane, only. The test specimen shall be conditioned in a controlled environment of 97 ±2°C and 97 ±2% relative humidity without condensate for a period of 28 days ±2 hours. To prevent condensate formation on the specimen, prior to removal of the specimen, the temperature and humidity of the chamber shall be reduced to room conditions. The specimen shall then be maintained at room temperature and 50% RH for 24 ±2 hours prior to testing in accordance with the applicable specification sheet.

5. PREPARATION FOR DELIVERY

5.1 Not applicable to this specification.

6. NOTES

6.1 Intended use. The insulating and jacketing compounds covered by this specification are intended for use in insulations and jackets respectively of electrical cables, cords, and wires.

6.2 Temperature rating. Typical temperature ratings for compounds covered by this specification are as follows. These values are not insured by the requirements of this specification, particularly for continuous operation. Acceptable periods of conditioning and low-temperature characteristics of the several materials covered under this specification may differ appreciably at the same minimum ambient temperatures. It is intended that the specification covering the completed cables, cords, or wires to which this specification is subsidiary, specify the conditioning period for the completed cables, cords, or wires, the required tests (cold bend, flexibility or others) and performance data defining limits of operation for samples of finished cables, cords, or wires at temperatures coinciding with the minimum temperature ratings, as applicable, indicated herein.

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COMPOUND	MINIMUM AMBIENT TEMP °C	MAXIMUM COPPER TEMP °C
PVC:		
Medium-low temperature	-40	+80
SBR:		
Medium-low temperature	-40	+75
Low temperature, heat-resistant	-55	+75
Unjacketed, medium-low temperature	-40	+60
Medium low-temperature, insulating and jacketing (IJS)	-40	+75*
Natural rubber:		
Low temperature	-55	+60
Rubber latex-chlorprene rubber latex, low temperature	-55	+85
Rubber latex:		
Low temperature	-55	+60
Low temperature, heat-resistant	-50	+80
Chloroprene rubber:		
Moderate-low temperature	-40	+85*
Low temperature, heat-resistant	-55	+85*
EPM and EPDM rubbers:		
Low temperature, heat resistant, ozone resistant, insulating	-55	+85
Low temperature, heat resistant, weather resistant, insulating and jacketing	-55	+90*
Low temperature, heat resistant, weather resistant, jacketing	-55	+90*
Polyethylene:		
Low temperature, heat resistant, insulating	-55	+80
Polyurethane:		
Low temperature, heat resistant, weather resistant, jacketing	-55	+85*

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COMPOUND	MINIMUM AMBIENT TEMP °C	MAXIMUM COPPER TEMP °C
Thermoplastics Elastomers		
Styrinic TPE-Low temperature, heat and ozone resistant, insulating and jacketing	-60	+85
Olefinic TPE-Low temperature weather resistant, insulating and jacketing	-60	+80

*Includes ambient temperature plus sun radiation.

6.3 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of applicable specification sheet.
c. Certification of compliance from source of manufacture (see 4.2.1).
d. Test results (see 4.4.3).

6.4 Test specimens. Specimens of insulation and jacketing are taken when all manufacturing processes are completed. Jackets (other than thermoplastic) shall be vulcanized prior to sampling of the jacket and/or insulation. Insulation specimens may be taken directly from the conductors when the conductors are to be furnished unjacketed.

6.5 Restriction on use of rubber latex. Rubber latex should not be used to insulate tinsel ribbon inasmuch as .5 percent free elemental sulfur, as permitted herein, will cause tarnishing of the tinsel ribbon, and thus result in noisy cords.

6.6 Restriction on use of thermoplastics elatomers. Thermoplastic elastomers may be incompatible with current method of termination of shrinking a boot onto the cable.

6.7 First article inspection. When a first article inspection is required, the item will be tested and should be a sample unit produced with equipment and procedures normally used in production. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the document's first article.

6.7.1 First article inspection on repeat orders. First article inspection will not be required from a supplier who has manufactured the same type of insulation and jacket materials, as applicable, on a previous order or contract within 2 years of the current order or contract, and which passed first article inspection at that time. Any changes made in the compound formulation for the type previously tested and approved shall require first article retest.

6.8 Revisions. Revision letters are not used to denote changes due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-3930E	Insulating and Jacketing Compounds, Electrical (For Cables, Cords, and Wires), General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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